



Performance Evaluation of Cloud Native Applications: A Systematic Mapping Study

Francisco Airtton Silva¹ · Fernando A. M. Trinta² · Michel S. Bonfim² · José Antonio F. de Macedo² · Paulo A. L. Rego² · Vinícius Lagrota³

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Abstract

Cloud-native applications leverages cloud environments for enhanced scalability, resilience, and agility. Using containerization technologies like Docker enables modular deployments. Despite their advantages, effectively evaluating the performance of these dynamic and distributed applications presents challenges. Cloud-native application performance is complex because diverse metrics—including response times, throughput, and resource utilization—are used across varied workloads. Organizations face hurdles in selecting appropriate evaluation tools, defining specific objectives, and gaining insights into microservices and orchestration. A more structured approach is essential to optimizing deployments and meeting performance expectations in cloud-native environments. This paper presents a systematic mapping study that examined 1158 initial papers and refined them to 32 primary studies. It meticulously analyzes key metrics, software tools, objectives, and contributions related to performance evaluation in cloud-native applications. Visual representations with bubble plots were used as cross-correlation analysis, revealing patterns and research gaps. This study provides insights that guide future research directions, advocating for methodological frameworks to enhance the assessment and optimization of cloud-native application performance.

Keywords Mapping study · Cloud-native applications · Performance · Metrics · Research contributions

1 Introduction

Cloud-native applications context. The projected growth rate for the global market size of cloud-native applications by 2028 is expected to reach 23.7%, amounting to approximately USD 17 billion [1]. In comparison, the market size stood

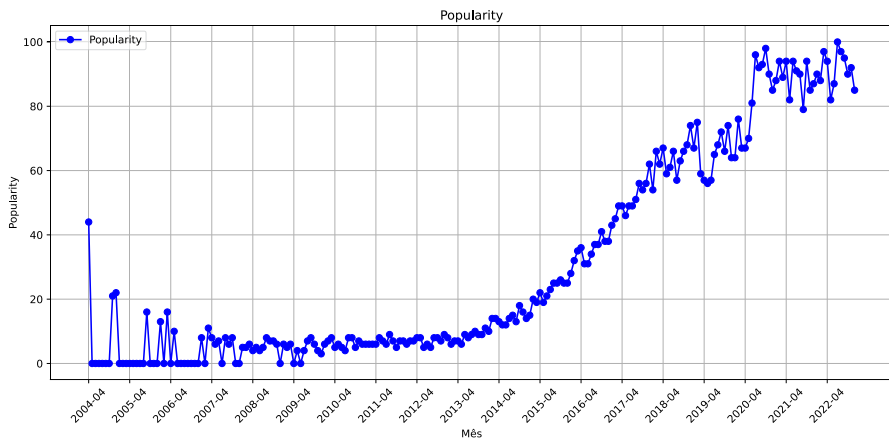


Fig. 1 Google trends regarding cloud-native expression in the last 20 years. Google Trends normalizes search data (0 - 100) to make comparisons between terms easier

at USD 5.9 billion in 2023. The popularity of cloud-native applications is illustrated by Google Trends data for the term “cloud-native” over the past 20 years (Fig. 1). Cloud-native applications represent a paradigm shift in software development, offering a modern approach to building and deploying applications that are optimized for cloud environments. They are built using a microservices architecture, where applications are decomposed into smaller, loosely coupled services that can be developed, deployed, and scaled independently [3]. Containerization technologies, such as Docker, are often used to package these services into lightweight and portable units, enabling consistent deployment across different environments. Moreover, orchestration tools such as Kubernetes automate container management [12]. Embracing cloud-native principles improves scalability and adaptability [10].

Performance evaluation ensures the effectiveness of cloud-native applications [19]. Cloud-native applications leverage microservices architecture, containerization, and orchestration tools to achieve scalability, resilience, and agility in cloud environments. However, the dynamic and distributed nature of these applications poses unique challenges for performance assessment [7]. Evaluating the performance of cloud-native applications involves analyzing various metrics such as response times, throughput, resource utilization, and scalability under different workload conditions. The importance of performance evaluation in this context cannot be overstated. It enables organizations to identify bottlenecks, optimize resource allocation, and ensure reliable and responsive user experiences [2]. Moreover, practical performance evaluation facilitates informed decision-making regarding infrastructure provisioning, cost optimization, and application design enhancements. As organizations increasingly rely on cloud-native architectures to meet the demands of modern digital businesses, understanding and optimizing the performance of these applications is essential for maintaining competitiveness and meeting customer expectations.

The research problem. In the context of cloud-native applications, conducting effective performance evaluations poses a challenge. This complexity arises from several factors, including the need to choose the right tools from a wide range of options, define clear objectives for each application, select relevant metrics from numerous possibilities, and determine the potential insights gained from the evaluation process. This challenge is exacerbated by the intricate nature of cloud-native applications, which are based on microservices, containerization, and orchestration. As organizations strive to leverage the scalability, resilience, and agility offered by cloud-native architectures, the lack of a structured approach to performance evaluation hinders their ability to optimize and assess their deployments. This research problem highlights the urgent need for a systematic mapping study to consolidate existing knowledge, identify aspects of performance evaluation, and suggest directions for future research and practical application.

The proposed solution. We conducted a systematic mapping study to explore metrics, software programs, objectives, and contributions related to the performance evaluation of cloud-native applications. We aimed to provide a thorough overview of the current research landscape in this area. Starting with 1158 articles and narrowing down through successive refinement stages to 32 primary studies for detailed analysis. We meticulously analyzed these studies, examining individual factors and their correlations. Utilizing bubble plots, we visually cross-correlated the results, enriching our analysis by illustrating relationships among different factors identified in the literature. This visual method not only facilitates the identification of patterns and trends but also provides deeper insights into the interactions within various aspects of cloud-native application performance evaluation. By integrating quantitative data with visual representations, our study improves understanding and offers a nuanced perspective on the evolving research landscape of cloud-native applications.

The subsequent sections are organized as follows. Section 2 delves into related works. Section 3 describes the methodology used. Section 4 presents classification-based findings. Section 5 discusses a categorization derived from intersecting classifications. Section 6 outlines identified challenges and future directions. Lastly, Sect. 7 concludes with recommendations for future research in this area.

2 Related Works

This section presents studies that have conducted work similar to ours. We identified thirteen articles that have performed literature reviews, explicitly focusing on systematic mapping studies, systematic literature reviews, or surveys. These reviews centered around various aspects related to cloud-native applications. Table 1 provides an overview of related works. We will discuss the table's contents by reviewing the primary columns sequentially from left to right.

Year of Publication Analysis of the publication year reveals a trend: the majority of studies emerged after the onset of the pandemic in 2020, with only two papers published prior to this period, indicating a current and growing interest in research on cloud-native applications. Specifically, eight out of thirteen papers were published in 2023, signaling a marked increase in scholarly attention to this topic during

Table 1 Secondary studies that focus on investigating literature regarding aspects of cloud-native applications

References	Year	Type	Context	Title
[10]	2017	Systematic Mapping Study	General	Understanding cloud-native applications after 10 years of cloud computing-a systematic mapping study
[13]	2019	Systematic Mapping Study	Architecture	A Systematic Mapping Study of Cloud-native Application Design and Engineering
[18]	2021	Survey	Network	Cloud-native network slicing using software-defined networking based multi-access edge computing: A survey
[14]	2022	Survey	Monetary Cost	A Survey on Billing Models for Cloud-Native Applications
[3]	2023	Systematic Literature Review	Architecture	Understanding the challenges and novel architectural models of multi-cloud native applications-a systematic literature review
[9]	2023	Systematic Mapping Study	Architecture	Toward the Observability of Cloud-Native Applications: The Overview of the State-of-the-Art
[16]	2023	Systematic Mapping Study	Security	Static-Analysis-Based Solutions to Security Challenges in Cloud-Native Systems: Systematic Mapping Study
[20]	2023	Survey	Security	Security in Cloud-Native Services: A Survey
[11]	2023	Systematic Mapping Study	Architecture	Multi-tenancy in Cloud-native Architecture: A Systematic Mapping Study
[21]	2023	Survey	Deployment	Cloud-native workload orchestration at the edge: A deployment review and future directions
[17]	2023	Systematic Mapping Study	Security	Access control design practice and solutions in cloud-native architecture: A systematic mapping study
[6]	2023	Survey	Network	A Survey on Resource Management for Cloud Native Mobile Computing: Opportunities and Challenges
[5]	2024	Survey	Architecture	Cloud-Native Computing A Survey From the Perspective of Services
This Paper	-	Systematic Mapping Study	Performance Evaluation	Performance Evaluation of Cloud Native Applications: A Systematic Mapping Study

that year. Furthermore, the publication of one paper in 2024 underscores sustained academic interest in the subject.

Type of Research *Systematic mapping studies, systematic literature reviews, and surveys* are methodologies commonly utilized in academic research to consolidate and analyze existing knowledge within a specific domain. A *systematic mapping study* categorizes and maps the breadth of research within a field, aiming to identify trends, gaps, and the scope of available literature. It provides a broad overview by synthesizing a wide range of studies and organizing them based on predefined criteria. In contrast, a *systematic literature review* focuses on synthesizing and analyzing in-depth findings from selected studies, with the objective of providing a deeper understanding of specific research questions or themes. It evaluates methodologies, findings, and theoretical contributions to uncover patterns or contradictions across the literature. *Surveys*, such as narrative reviews or scoping reviews, gather insights from existing literature in a less structured manner. They aim to provide a broad understanding of a topic by summarizing key findings and highlighting trends or issues. Each methodology contributes uniquely to advancing knowledge within their respective fields, collectively enhancing the understanding and applicability of research findings.

We have adopted a systematic mapping study because, in the context of performance evaluation, where definitions, metrics, and methodologies can vary widely, a systematic mapping study ensures that all relevant studies are considered, providing a holistic view of the current state of knowledge. This approach systematically identifies and organizes existing research, clarifying what has been studied and revealing gaps and inconsistencies in the literature. Moreover, this study's systematic design enhances transparency, reproducibility, and the reliability of findings, making it an ideal method for navigating the complexities and uncertainties inherent in evaluating cloud-native applications' performance.

Context

Most of the papers were classified into three categories: architecture, security, and network. We will now discuss these categories specifically.

Architecture. Five recent papers have contributed to understanding various aspects of cloud-native applications in terms of architecture [3, 5, 9, 11, 13]. Deng et al. [5] conducted a survey focusing on issues throughout the cloud-native application lifecycle from a service perspective. Alonso et al. [3] characterized the concept of multi-cloud and defined the characteristics of multi-cloud native applications while also analyzing the current trends and challenges in multi-cloud environments. Odun et al. [13] conducted a systematic mapping study to explore the designs of cloud-native applications and shed light on their frequency in research. Kosinska et al. [9] assessed observability in cloud-native applications through a systematic mapping study, identifying motivations, research areas, and future trends in this domain. Matthew et al. [11] identified trends, challenges, and opportunities related to multi-tenancy within cloud-native architecture, exploring both improvements and challenges in this area.

Security. Three papers have explored critical aspects of security and access control in cloud-native environments [16, 17, 20]. Theodor et al. [20] explored various security facets specific to cloud-native services, identifying key features and

proposing corresponding cybersecurity solutions. Rahaman-a et al. [16] conducted a structured survey focusing on defense mechanisms employed in cloud-native systems, providing insights into the diverse approaches to enhancing system security. Similarly, Rahaman-b et al. [17] analyzed existing access control mechanisms within cloud-native architecture, categorizing access control patterns and offering practical guidance for developers.

Network. Two papers have addressed the network category [6, 18]. Shah et al. [18] investigated network slicing, emphasizing their application in 5 G services and exploring recent technological advancements that enable end-to-end (E2E) network slicing. Similarly, Huang et al. [6] conducted a survey on resource management within cloud-native mobile computing, with a specific focus on challenges related to network slicing and containerization resource management.

Others. The remaining three papers were classified into other particular contexts [10, 14, 21]. Vano et al. [21] reviewed the orchestration of cloud-native workloads at the edge, emphasizing challenges related to containerization, exploring pertinent tools, and outlining future directions in edge computing. Meanwhile, Kratzke et al. [10] contributed general insights by defining cloud-native applications through systematic mapping outcomes while also summarizing current research trends and engineering approaches within this domain. Furthermore, Paredes et al. [14] explored micro-billing models for Cloud-Native Applications, identifying potential opportunities and proposing a theoretical framework to enhance billing efficiency in cloud-native environments.

None of the reviewed papers focused on the performance aspect, highlighting a gap in the existing literature. As a result, our study addresses distinct research questions from those explored in prior work. Specifically, we conducted a systematic mapping study to map metrics, tools, objectives, and contributions pertaining to the performance evaluation of cloud-native applications. This approach aims to provide an overview of the current state of research in this area.

3 Methodology

This section presents the methodology of this work. A systematic mapping study was applied to understand the topic of interest. A systematic mapping study performs “*a broad review of primary studies in a specific topic area that aims to identify what evidence is available on the topic.*” [8]. Systematic mapping studies offer several advantages, including providing an overview of a research area, identifying research gaps, and facilitating evidence-based decision-making. By systematically mapping existing literature, researchers can gain insights into the current state of knowledge, highlight areas that require further investigation, and inform future research directions. These studies help synthesize a large volume of information, enabling researchers to identify trends, patterns, and inconsistencies across studies. This is valuable for policymakers, practitioners, and researchers alike. Additionally, systematic mapping studies help reduce duplication of research efforts, enhance resource allocation efficiency, and promote collaboration among researchers in similar domains. We have adopted the systematic mapping study process proposed by

Fig. 2 Steps of the systematic mapping study conducted in this research

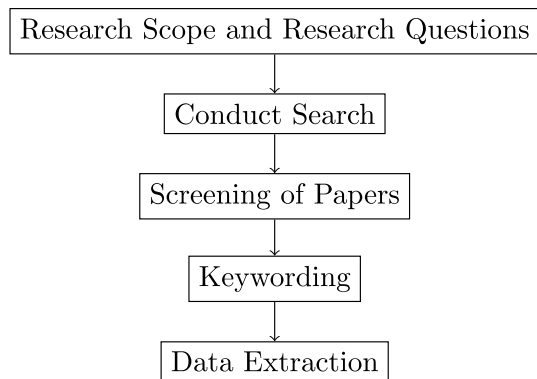


Table 2 List of research questions

Question IDs	Description
RQ01	What are the metrics evaluated in this topic?
RQ02	What main software programs were used or evaluated in this topic?
RQ03	What are the specific paper objectives ?
RQ04	What are the main research contributions ?

Peterson et al. [15]. This process is visualized in Fig. 2 and explained in the following subsections.

3.1 Research Scope and Research Questions

Understanding the methods for evaluating the performance of cloud-native applications is crucial in today's digital landscape, characterized by migration to cloud-based infrastructure. As organizations increasingly rely on cloud-native architectures to deploy and scale their applications, assessing their performance becomes paramount. The rapid evolution of cloud technologies introduces complexities in performance evaluation, prompting researchers and practitioners alike to explore novel methodologies and tools to address challenges effectively. By delving into the current state-of-the-art, we can uncover approaches, metrics, and frameworks employed to evaluate the performance of cloud-native applications. This exploration observes existing methodologies and serves as a foundation for identifying gaps in the literature, paving the way for future research directions aimed at enhancing performance evaluation practices in cloud-native computing. Based on these arguments, this paper tries to answer the following question:

What is the state-of-the-art related to performance evaluation of cloud-native applications?

Many questions can be derived from such a general question. However, based on the expectations of the specialists and practitioners consulted, we have delimited our efforts to answer four questions presented in Table 2.

RQ01. The state-of-the-art in performance evaluation of cloud-native applications involves assessing various metrics to ensure optimal functionality. Metrics such as response time, resource utilization, service time, and throughput are essential in evaluating the performance of cloud-native systems [4, 22]. These metrics help in understanding the efficiency, reliability, and scalability of cloud-native applications, enabling researchers and practitioners to identify areas for improvement and optimization. By analyzing these metrics, it becomes possible to enhance the overall performance of cloud-native systems, ensuring they meet the required service level objectives and deliver a seamless user experience. However, it is interesting to know how the majority of metrics are used in this area and how they can be combined with other aspects.

RQ02. Understanding the primary software programs employed or assessed in the performance evaluation of cloud-native applications holds importance for various reasons.

The software Kubernetes and AWS tools are highly used tools in this area. However, there are many others. This question provides invaluable insights into the technological ecosystem of cloud-native environments, unveiling the predominant tools and platforms utilized by both practitioners and researchers. This understanding empowers stakeholders to make informed decisions regarding tool selection and infrastructure choices for their performance evaluation endeavors. Identifying the leading software programs offers a nuanced understanding of their capabilities, limitations, and compatibility with diverse cloud-native architectures. Such knowledge facilitates the selection of appropriate tools tailored to specific evaluation requirements, ensuring optimal performance assessment outcomes. This exploration not only fosters continuous improvement in evaluation methodologies but also drives the development of more sophisticated and practical tools to address evolving performance challenges in cloud-native environments.

RQ03. Understanding the specific objective of cloud-native performance evaluation is crucial for several reasons. Firstly, cloud-native architectures introduce a myriad of unique objectives, such as orchestration, resource management, optimization, and benchmarking. Focusing on specific objectives allows researchers and practitioners to tailor their evaluation methodologies to address these nuances effectively. Pinpointing and defining these specific elements provide emphasis and clarity to performance evaluation processes. Exploring narrowly targeted areas promotes a deeper understanding of the complexities of cloud-native settings. This greater understanding may result in the creation of specialized instruments, measurements, and benchmarks precisely calibrated to evaluate performance in these particular areas.

RQ04. Understanding the main research contributions in cloud-native performance evaluation, advances our understanding and capabilities in optimizing cloud-native applications. By identifying key areas such as evaluation, method, tool, and algorithm, we uncover the diverse facets of performance assessment within cloud-native environments. Highlighting the leading research contributions in cloud-native performance evaluation underscores the importance of ongoing research in this area. Exploring these contributions allows us to

identify trends, address existing challenges, and propel the field forward, ultimately improving the performance and scalability of cloud-native applications in real-world scenarios.

3.2 Conduct Search (for primary studies)

We conducted our search for studies in the electronic sources listed in Table 3. We searched for contributions published with the three terms “cloud,” “native,” and “performance” in their title, abstract, or keywords. However, we did not perform a full-text search. Authors should intentionally consider the cloud-native performance evaluation aspect so much that they explicitly mention it in the title, abstract, or as a keyword.

3.3 Screening of Papers (for inclusion and exclusion)

Inclusion and exclusion criteria are indispensable elements of mapping studies, serving as a navigational compass that guides researchers through the vast sea of literature. These criteria ensure precision in research scope by delineating the boundaries of relevance, allowing scholars to focus exclusively on literature that directly addresses their research questions and objectives. They minimize bias by providing transparent guidelines for selection and optimize resource utilization by efficiently allocating time and effort to synthesizing high-quality evidence. Moreover, adherence to these criteria improves the reproducibility and transparency of mapping studies, enabling other researchers to validate findings and build upon existing knowledge. In essence, inclusion and exclusion criteria are the cornerstones of methodological rigor in mapping studies. They facilitate the synthesis of meaningful insights that contribute to the advancement of scholarly discourse and understanding in various fields. Table 4 presents the inclusion and exclusion criteria for this work. Table 5 shows the 32 selected studies.

Table 3 Search strings specific for each search engine

Search Engines	Quantity	Search String
IEEE Xplore	422	(“Abstract”:cloud native performance)
ACM Digital Library	192	[Abstract: cloud] AND [Abstract: native] AND [Abstract: performance]
Google Scholar	355	Search for “cloud native performance”, then click on Sort by date - Abstracts
ScienceDirect	75	Title, abstract, keywords: cloud AND native AND performance
SpringerLink	114	cloud AND native AND performance
TOTAL	1158	

Table 4 Inclusion and exclusion criteria

Inclusion	Exclusion
• The abstract explicitly mentions the performance evaluation of cloud-native applications. • From the abstract, we can deduce that the focus of the paper contributes to the chosen research focus. • Abstract/keywords include key terms, and it is clear from the abstract that a contribution towards cloud-native applications and their evaluation is made	• The paper lies outside the performance evaluation of cloud-native applications context. • Performance evaluation of cloud-native applications is not part of the contributions of the paper; the terms are only mentioned in the general introductory sentences of the abstract. • Literature not peer-reviewed and only in the form of abstract, blog, or presentation are excluded

3.4 Keywording

We have read all the selected papers and extracted the keywords that could answer one of the four research questions. We have found 48 words (or expressions), which were grouped (and ordered by importance) and presented in Table 6. It is essential to highlight that the words were unified when they appeared with similar meanings, as suggested by Peterson et al. [15]. We have found 20 metrics, 17 software programs, 7 objectives, and 4 main research contributions.

3.5 Data extraction

In this step, we associate the keywords to each paper, producing the results listed in Table 7. This table allows the reader to access the specific paper that addresses the desired characteristics. However, we will analyze each category in two ways: isolated and associated with each other.

4 Research Answers Analyzed Individually

In this section, we examine each question's response individually, aiming to extract the most and least utilized options while providing insightful commentary. We visualize the frequency of occurrence for each option using bar plots, facilitating clear comparisons. Throughout our analysis, we focus on discussing the most adopted items, providing insight into their relevance within the study's context.

4.1 RQ01 - Metrics

Figure 3 shows the used metrics organized by number of occurrences. The first six metrics are intrinsically related to performance, and therefore, we focus on discussing them. The three most used metrics (latency, response time, and service time) can be grouped as measures related to time. These metrics are correlated metrics that collectively provide insight into the performance of systems and networks. Latency represents the time taken for data to traverse between

Table 5 Primary studies detailed as references

ID	Title
PS01	“Performance evaluation of two service recovery strategies in cloud-native radio access networks,” Shunmugapriya Ramanathan et al., 2019 21st International Conference on Transparent Optical Networks (ICTON), IEEE, 2019
PS02	“Towards latency-sensitive cloud-native applications: A performance study on AWS,” István Pelle et al., 2019 IEEE 12th International Conference on Cloud Computing (CLOUD), IEEE, 2019
PS03	“X-containers: Breaking down barriers to improve performance and isolation of cloud-native containers,” Zhiming Shen et al., Proceedings of the Twenty-Fourth International Conference on Architectural Support for Programming Languages and Operating Systems, 2019
PS04	“Control-over-the-cloud: A performance study for cloud-native, critical control systems,” Per Skarin et al., 2020 IEEE/ACM 13th International Conference on Utility and Cloud Computing (UCC), IEEE, 2020
PS05	“Evaluations of network performance enhancement on cloud-native network function,” Yong-Xuan Huang, Jerry Chou, Proceedings of the 2021 on Systems and Network Telemetry and Analytics, 2020, 3-8
PS06	“PRESTO: a latency-aware power-capping orchestrator for cloud-native microservices,” Rolando Brondolin, Marco D. Santambrogio, 2020 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), IEEE, 2020
PS08	“Validation and Benchmarking of CNFs in OSM for pure Cloud Native applications in 5 G and beyond,” Adrián Pino et al., 2021 International Conference on Computer Communications and Networks (ICCCN), IEEE, 2021
PS09	“CloudNPlay: Resource optimization for a cloud-native gaming system,” Angelus Wibowo, Ta Nguyen Binh Duong, 2021 IEEE 30th International Conference on Enabling Technologies: Infrastructure for Collaborative Enterprises (WETICE), IEEE, 2021
PS10	“High-performance software load balancer for cloud-native architecture,” Jung-Bok Lee et al., IEEE Access, v. 9, p. 123704-123716, 2021
PS11	“Performance evaluation and improvement of a cloud-native data analysis system application,” Chia-Chuan Chuang, Yu-Chin Tsai, 2021 international conference on electronic communications, internet of things and big data (ICEIB), IEEE, 2021
PS07	“SRv6-PM: A cloud-native architecture for performance monitoring of SRv6 networks,” Pierpaolo Loreti et al., IEEE Transactions on Network and Service Management, v. 18, n. 1, p. 611-626, 2021
PS12	“A configurable method for benchmarking scalability of cloud-native applications,” Sören Henning, Wilhelm Hasselbring, Empirical Software Engineering, v. 27, n. 6, p. 143, 2022
PS13	“Applicability of cloud native-based healthcare monitoring platform (cn-hmp) in older adult facilities,” Abu Bony Amin et al., 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2022
PS14	“Availability and latency aware deployment of cloud native edge slices,” Sagar Arora, Adlen Ksentini, Christian Bonnet, GLOBECOM 2022-2022 IEEE Global Communications Conference, IEEE, 2022
PS15	“Democratizing the computational environmental marine data science: using the high-performance cloud-native computing for inert transport and diffusion lagrangian modelling,” Gennaro Mellone et al., 2022 IEEE International Workshop on Metrology for the Sea; Learning to Measure Sea Health Parameters (MetroSea), IEEE, 2022
PS16	“Latency and network aware placement for cloud-native 5 G/6 G services,” Kiranpreet Kaur et al., 2022 IEEE 19th Annual Consumer Communications & Networking Conference (CCNC), IEEE, 2022
PS17	“Microservices configurations and the impact on the performance in cloud-native environments,” Mohamed Mekki, Nassima Toumi, Adlen Ksentini, 2022 IEEE 47th Conference on Local Computer Networks (LCN), IEEE, 2022

Table 5 (continued)

ID	Title
PS18	"Novel Abstraction and Offload Mechanisms for High Performance Cloud-native Distributed Object Stores," Anoop Reddy et al., 2022 IEEE International Conference on Cloud Engineering (IC2E), IEEE, 2022
PS19	"Optimizing Performance and Resource Consumption of Cloud-Native Logging Application Stacks," Gergő Csáti, István Pelle, László Toka, NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium, IEEE, 2022
PS20	"Performance Evaluation of Resource Management Schemes for Cloud Native Platforms with Computing Containers," Yuqi Fu et al., 2022 IEEE International Performance, Computing, and Communications Conference (IPCCC), IEEE, 2022
PS21	"Scalable Digital Pathology Platform Over Standard Cloud Native Technologies," Tibério Baptista et al., 2022 IEEE Symposium on Computers and Communications (ISCC), IEEE, 2022
PS22	"Toward Performance Efficient UAV Task Scheduling in Cloud Native Edge," Siqi Huang, Deze Zeng, Zhihao Qu, GLOBECOM 2022-2022 IEEE Global Communications Conference, IEEE, 2022
PS23	"A Comprehensive Performance Analysis of Stream Processing with Kafka in Cloud Native Deployments for IoT Use-cases," István Pelle et al., NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium, IEEE, 2023
PS24	"CDSBen: Benchmarking the Performance of Storage Services in Cloud-Native Database System at ByteDance," Jiashu Zhang et al., Proceedings of the VLDB Endowment, v. 16, n. 12, p. 3584-3596, 2023
PS25	"Cloud-native-bench: an extensible benchmarking framework to streamline cloud," Michiel Van Kenhove et al., Proceedings of IEEE CloudNet2024, the IEEE International Conference on Cloud Networking, 2023
PS26	"High-Level Data Abstraction and Elastic Data Caching for Data-Intensive AI Applications on Cloud-Native Platforms," Rong Gu et al., IEEE Transactions on Parallel and Distributed Systems, 2023
PS27	"PerfSim: A performance simulator for cloud-native microservice chains," Michel Gokan Khan et al., IEEE Transactions on Cloud Computing, v. 11, n. 2, p. 1395-1413, 2021
PS28	"Zero-cost in-depth enforcement of network policies for low-latency cloud-native systems," Gerald Budigiri et al., 2023 IEEE 16th International Conference on Cloud Computing (CLOUD), IEEE, 2023
PS29	"Disambiguating Performance Anomalies from Workload Changes in Cloud-Native Applications," Alexandru Baluta et al., Proceedings of the 15th ACM/SPEC International Conference on Performance Engineering, 2024
PS30	"Lightpool: A nvme-of-based high-performance and lightweight storage pool architecture for cloud-native distributed database," Jiexiong Xu et al., 2024 IEEE International Symposium on High-Performance Computer Architecture (HPCA), IEEE, 2024
PS31	"Performance Optimization Across the Edge-Cloud Continuum: A Multi-agent Rollout Approach for Cloud-Native Application Workload Placement," Polyzois Soumplis et al., SN Computer Science, v. 5, n. 3, p. 318, 2024
PS32	"Towards High-Performance Data Loading in Cloud-Native Deep Learning Systems," Zhuangwei Kang et al., 2024 16th International Conference on COMMunication Systems & NETWORKS (COMSNETS), IEEE, 2024

endpoints. Response time encompasses latency along with the time spent waiting in queues and processing the request on the server. Within response time, service time isolates the duration spent exclusively on processing the request by the server, excluding network-related delays. These three metrics offer a view of

Table 6 Unified keywords

Classification	Keywords
Metrics	latency, response time, service time, throughput, resource utilization, scalability, packet loss, power consumption, monetary cost, availability, number of successful executions, frame rates, instantiation time, Kafka KPIs, IO rate, simulation accuracy, simulation speed, precision, recall, resource allocation success
Software	Kubernetes, Some AWS tool, Prometheus, Docker, Proprietary tool, Tensorflow, Pytorch, Spark, Grafana, gRPC, Big Data tools, OSM-OpenStack, Kafka, Fluid, Theodolite, PRONet Orchestrator, libOS, Calico
Objective	orchestration, resource management, optimization, benchmarking, monitoring, storage, network performance
Main Research Contribution	evaluation, method, tool, algorithm

system and network performance and are easily perceived by the end user [PS08] [PS01] [PS04].

Throughput is the fourth metric and gives a more holistic view of infrastructure performance. It provides information on the application's ability to handle increasing workloads, ensuring optimal resource utilization (the fifth metric) while maintaining responsiveness. Moreover, in the context of microservices, throughput measurements aid in identifying performance bottlenecks and optimizing service interactions. Throughput performance was extensively analyzed in the research papers provided. For instance, the papers discussed how increasing computation performance benefits accessing the same key with a few concurrent functions [PS02], the performance differences between loopback and Load Balancer Node (LBN) in terms of bps rate and pps rate [PS09], the system call throughput improvements with X-Containers compared to Docker and Clear Containers [PS03], the impact of gradient computation time on throughput for ImageNet and LibriSpeech datasets [PS32], and the mean throughput of different web servers under specific resource restrictions [PS25]. The sixth metric (packet loss) is primarily used when the evaluator observes failure aspects, mainly during stress tests. Packet loss is sometimes hard to reach, considering powerful computing equipment, but it was observed in four papers.

4.2 RQ02 - Software Programs

Figure 4 shows the used software programs organized by the number of occurrences. Kubernetes was extensively utilized in the research papers studied. The studies implemented Kubernetes for various purposes, such as managing applications in containers, conducting performance analyses, and automating benchmark deployments. For instance, Kristiani et al. [PS02] utilized Kubernetes to implement an edge-computing architecture and distributed cloud applications. Additionally, the studies highlighted the importance of Kubernetes operators like Theodolite to benchmark scalability in microservices and ImageJockey to evaluate container image performance [PS09]. Furthermore, the research emphasized the significance of Kubernetes in orchestrating cloud-native applications, enforcing resource

Table 7 Categorization of each paper

ID	Metrics	Software	Objective	Main Research Contribution
PS01	throughput and resource utilization	PRONet Orchestrator	orchestration	evaluation
PS02	throughput, service time, number of successful executions	Some AWS tools	orchestration	evaluation
PS03	latency, packet loss and throughput	libOS	optimization	method
PS04	latency and service time	Some AWS tools	benchmarking	evaluation
PS05	latency, packet loss, and throughput	Calico and Kubernetes	network performance	evaluation
PS06	power consumption, packet loss and latency	Docker and Kubernetes	orchestration	tool
PS08	latency, frame rates, response time, monetary cost	Some AWS tools	resource management	tool
PS09	throughput	Kubernetes	orchestration	evaluation
PS10	resource utilization and response time	Kubernetes, and Docker	resource management	evaluation
PS11	throughput	gRPC and Big Data tools	monitoring	method
PS07	latency and power consumption, resource utilization, instantiation time	Kubernetes and OSM-OpenStack	benchmarking	evaluation
PS12	scalability	Kubernetes	benchmarking	method
PS13	response time, packet loss, and latency	Proprietary tool	monitoring	evaluation
PS14	availability, latency	Kubernetes	orchestration	evaluation
PS15	response time, throughput, service time, scalability, and resource utilization	Kubernetes and Prometheus	orchestration	method
PS16	latency, service time	Kubernetes	optimization	algorithm
PS17	service time	Kubernetes	resource management	evaluation
PS18	response time, throughput, service time, scalability, and resource utilization	Some AWS tools	storage	evaluation
PS19	response time, throughput, and scalability	Kubernetes and Some AWS tools	monitoring	evaluation
PS20	service time, resource utilization and response time	Some AWS tools, Docker, Kubernetes, Tensorflow, Pytorch, and Spark	resource management	evaluation
PS21	response time and throughput	Kubernetes	resource management	method
PS22	service time, availability and latency	Some AWS tools	optimization	algorithm

Table 7 (continued)

ID	Metrics	Software	Objective	Main Research Contribution
PS23	response time and Kafka KPIs	Kafka	benchmarking	evaluation
PS24	latency, IO rate	Proprietary tool	storage	tool
PS25	scalability	Kubernetes	benchmarking	tool
PS26	response time and throughput	Fluid	orchestration	method
PS27	simulation accuracy, simulation speed, response time	Kubernetes	orchestration	tool
PS28	latency, throughput, response time, resource utilization	Kubernetes	resource management	algorithm
PS29	precision, recall	Theodolite and Kubernetes	resource management	method
PS30	resource utilization, latency, and throughput	Kubernetes	storage	method
PS31	resource allocation success, service time, latency, resource utilization, monetary cost	Proprietary tool	optimization	algorithm
PS32	throughput	Kubernetes	optimization	method

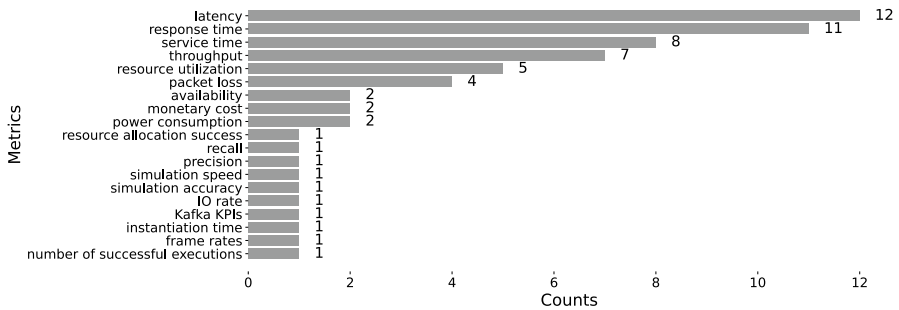


Fig. 3 Primary studies quantified by metrics

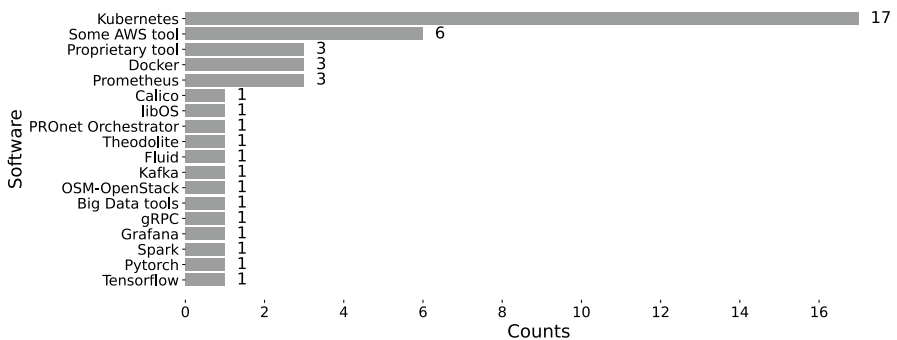


Fig. 4 Primary studies quantified by software programs

requests and limits, and effectively managing resource consumption [PS05]. In summary, Kubernetes facilitates efficient cloud architectures, automating benchmarking processes, and improving the performance of containerized applications across various studies.

In the second position resides Some AWS tools. AWS tools were extensively utilized for various purposes. The studies involved performance benchmarking of AWS Lambda and ECS across different regions, using tools like CloudWatch Logs to monitor hardware configurations [PS02]. Additionally, the papers discussed the implementation of a cloud-native deep learning system that leverages AWS S3 for data storage and access, showcasing the integration of AWS services within the system [PS27]. The analysis of cloud-native architectures and load-balancing mechanisms highlighted the importance of AWS tools in optimizing performance and scalability within cloud environments [PS09] [PS32]. Overall, the research papers demonstrated the diverse applications and benefits of utilizing AWS tools in enhancing cloud services and system functionalities. Three papers proposed proprietary tools, with Docker being utilized in only three papers despite its use in microservices contexts. Prometheus is featured prominently in research papers and has been extensively applied for diverse monitoring and management purposes. In one study [PS21], Prometheus was integrated into the proposed architecture using the Prometheus Stack Helm Package, connecting with the K8s' API and producing metrics

about nodes and objects in the K8S environment. Additionally, Prometheus was leveraged as an open-source monitoring solution, integrating with multiple metrics exporters to monitor applications and their environments alongside application metrics averages output in httpperf logs [PS29]. Additionally, another paper [PS25] emphasized the extension of the metrics server through the Prometheus Adapter Helm Package component. This enhancement enables Prometheus to query the deployment for custom metrics, facilitating precise system scaling based on metrics like concurrent DICOM associations and process memory usage.

4.3 RQ03 - Objectives

Figure 5 shows the objectives organized by number of occurrences. Resource management and orchestration are important in the mapping study on cloud-native performance evaluation due to their fundamental roles in ensuring the efficient operation and scalability of cloud-native applications. With the dynamic nature of cloud environments and the proliferation of microservices architectures, effective resource management is essential for optimizing resource allocation, maximizing utilization, and minimizing costs. Similarly, orchestration plays a critical role in automating the deployment, scaling, and management of application components, enabling coordination and efficient resource utilization across distributed environments. Container orchestrators like Kubernetes play a vital role in automating the deployment and management of containerized components within clusters, enabling features such as automatic container placement, replication, failover recovery, and scaling [PS21] [PS09]. Additionally, optimizing application performance while managing power consumption is essential, especially considering the non-linear relationship between server performance and energy usage, emphasizing the need for power-capping orchestrators to maintain workload performance near predefined SLAs [PS06]. Effective resource management and orchestration not only enhance software reliability and availability but also enable the creation, removal, and scaling of applications to meet varying traffic demands, ultimately ensuring the reliable and scalable delivery of cloud services [PS20].

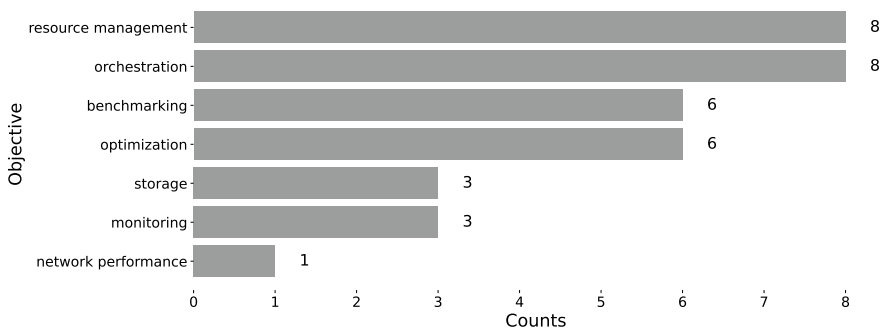


Fig. 5 Primary studies quantified by objectives

Benchmarking also stands out as the primary objective. Benchmarking allows for the comparison of different configurations, architectures, and technologies, providing valuable insights into performance variations and identifying best practices. By benchmarking cloud-native applications, researchers can assess their performance under various conditions and workload scenarios, facilitating informed decision-making and optimization efforts. Benchmarking in empirical software engineering serves to compare and rank different software under test, evaluate new methods or tools against established standards, and repeat previous experiments [PS12]. The complexity of cloud environments, with various storage options and cache systems, highlights the need for benchmarking to ensure optimal performance in data-intensive applications like deep learning training jobs [PS26]. Automated benchmarking tools like Cloud-Native-Bench streamline the process by monitoring benchmark states, automating consecutive executions, and facilitating result collection and analysis, enhancing the ease of use and efficiency of benchmarking workflows [PS25]. Benchmarking not only aids in understanding the scalability and performance of distributed systems but also enables users to make informed decisions based on specific business needs, ensuring the practical evaluation of different technologies in cloud environments.

Furthermore, optimization emerges as a central objective. In the context of cloud-native applications, optimization refers to the systematic process of fine-tuning various aspects of application design, infrastructure configuration, and resource utilization to improve performance, scalability, efficiency, and cost-effectiveness. Optimization involves identifying and addressing bottlenecks, inefficiencies, and areas for improvement within the application architecture and deployment environment. Optimization efforts include optimizing resource allocation and utilization, optimizing data storage and retrieval mechanisms, tuning networking configurations, and optimizing application code and algorithms. The goal of optimization in cloud-native applications is to maximize the utilization of available resources, minimize response times, enhance scalability, and ultimately improve the overall quality of service provided to users while minimizing operational costs. For instance, optimizing performance and resource utilization during deep learning training is essential, given the complexity of managing diverse storage systems and cache configurations in cloud environments [PS19] [PS26]. In this case, the motivation for optimization stems from the need to efficiently handle massive datasets and specialized computing accelerators in deep learning workloads, which are both data-hungry and computation-intensive.

4.4 RQ04 - Main Research Contributions

Figure 6 shows the main research contributions organized by a number of occurrences. Performance evaluation serves as the cornerstone contribution in the mapping study on cloud-native performance evaluation due to its inherent significance in assessing the effectiveness, reliability, and scalability of cloud-native applications. The mapping study focuses on performance evaluation and aims to provide a overview of existing methodologies, metrics, and tools for assessing the performance of

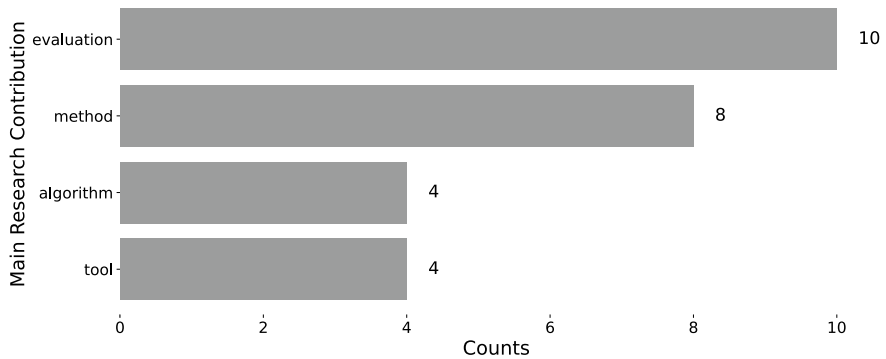


Fig. 6 Primary studies, quantified by main research contribution

cloud-native applications. The majority of the papers focused on performance evaluation of existing or proposed platforms. For instance, the evaluation of container platforms like Docker Swarm and Kubernetes on various setups was a vital aspect of the research in [PS20]. Similarly, other papers compared Management and Orchestration (MANO) platforms, such as OSM and ONAP, in terms of operational and functional metrics without introducing new tools [PS07]. Additionally, experimental evaluations were conducted on stream processing engines in Kubernetes clusters across different cloud environments to assess benchmarking results and execution times, emphasizing performance evaluation over tool development [PS10].

As cloud-native architectures continue to evolve, novel methods are essential for addressing the unique challenges associated with evaluating and optimizing the performance of these complex distributed systems. Several papers have proposed methods and platforms in the realm of cloud-native technologies and data science. These include the introduction of X-Containers and SRv6-PM for efficient deployment and management of distributed applications [PS13] [PS20], as well as the development of a Scalable Digital Pathology Platform utilizing standard cloud-native technologies to enhance medical imaging practices and enable remote diagnostics and telemedicine [PS21]. Additionally, the democratization of computational environmental marine data science has been addressed, showcasing advances in cloud-based architectures for demanding imaging scenarios with high-resolution images, ensuring scalability, data replication, and auto-scaling mechanisms for improved query throughput and reduced retrieval times [PS12].

5 Research Answers Analyzed Associatively

This section elucidates the frequency of studies based on the interrelationship between two distinct criteria, displaying the number of studies at the intersections of the x and y axes within bubble plots. This multidimensional analysis, considering various criteria simultaneously, helps identify prevalent research domains and

potential gaps for future exploration. Therefore, we will focus on these two aspects, drawing red and green rectangles to discussed examples of prevalent research domains and potential gaps, respectively. The color red indicates that the topic may have been overexplored, and green indicates that further research may be necessary.

5.1 Metrics x Main Research Contribution

Figure 7 presents the relationship between the metrics and the main research contribution.

Prevalent Research Domains The biggest bubble is the combination of throughput with the method. Throughput is closely tied to papers that introduce novel methods for benchmarking scalability in cloud-native applications [PS10][PS12][PS30][PS20]. [PS12] emphasizes the importance of scalability and performance in handling large datasets efficiently. [PS30] focuses on enhancing throughput

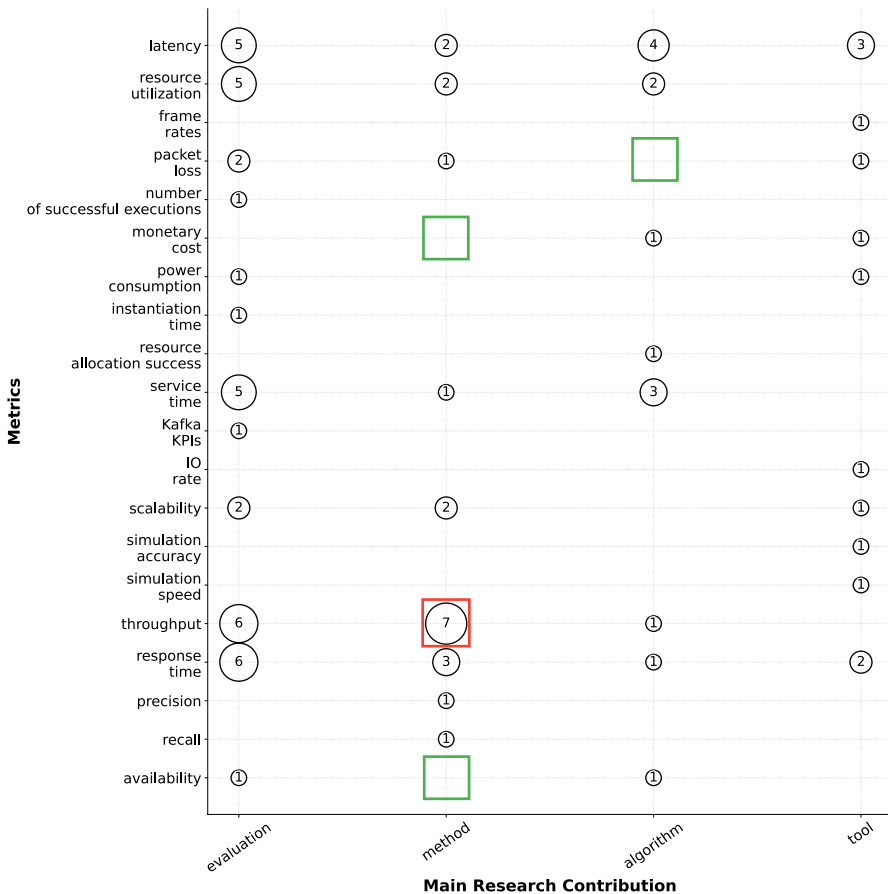


Fig. 7 Combination between metrics and main research contribution

by leveraging cloud-native technologies for efficient data processing and analysis. [PS20] also contributes to improving throughput by implementing advanced data abstraction and caching mechanisms for AI workloads on cloud-native environments. These papers collectively highlight the importance of methodological advancements in achieving high throughput in cloud-native applications.

Potential Gaps In the list of metrics discussed earlier, three metrics appeared only twice: packet loss, availability, and monetary cost. Despite their importance, these metrics have yet to be extensively explored. Deducing from the bubble plot, packet loss has a gap in the papers proposing algorithms. Tools like iperf accurately measure packet loss, indicating the reliability of counting systems in monitoring losses [PS11].

Availability offers a gap in method's and tool's papers. Availability is an essential aspect in various domains, such as telecommunication services, cloud-native systems, and object storage solutions. In the context of cloud-native systems, ensuring high availability is essential to minimize downtime and guarantee active microservices, with the goal of achieving 99.999% uptime [PS03]. Hot upgrades and hot migration in storage services increase availability, which prevents interruptions during component upgrades and data migration, ensuring continuous service availability [PS30]. Finally, the monetary cost has a gap in evaluation and method papers. When aiming to minimize monetary costs, the preference shifts towards utilizing cost-effective cloud nodes. Networking costs contribute to the total cost, especially when transferring data to the cloud. In contrast, prioritizing latency minimization through edge resources results in the processing cost of edge resources being the primary factor influencing the total monetary cost, with networking costs playing a minor role [PS31].

5.2 Software x Main Research Contribution

Figure 8 presents the relationship between software and main research contribution.

Prevalent Research Domains Among the many research efforts, eight papers illuminate the dynamics of application performance evaluation in Kubernetes-based environments. These studies explore various facets, ranging from container orchestration strategies to resource management techniques, all aimed at optimizing efficiency and scalability in cloud-native ecosystems. Through experimentation and analysis, these papers contribute to the collective understanding of how Kubernetes functions as a framework in modern cloud computing landscapes. Kubernetes is extensively utilized in performance evaluation studies, as evidenced by various research papers. The evaluation of resource management schemes for cloud-native platforms with computing containers highlights the importance of Kubernetes in optimizing performance [PS20]. Furthermore, the study on optimizing performance and resource consumption of cloud-native logging application stacks showcases how Kubernetes enhances efficiency and scalability in such environments [PS12]. Overall, these papers collectively demonstrate Kubernetes' significance in driving performance improvements and resource optimization in cloud-native applications.

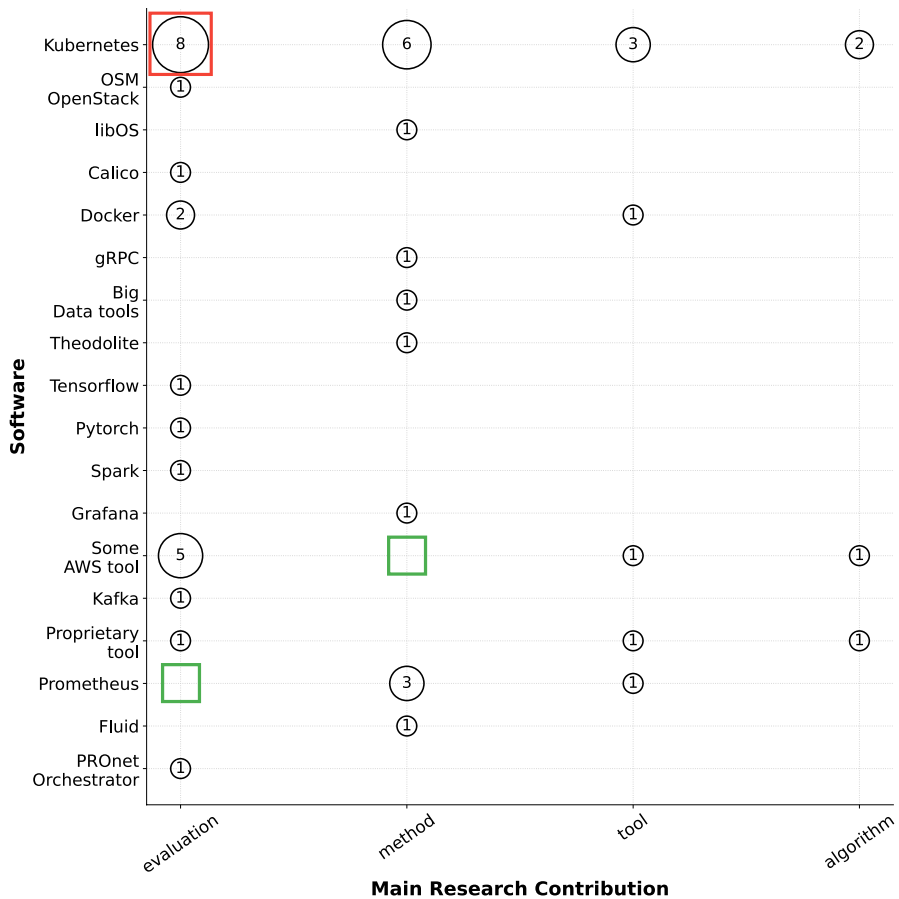


Fig. 8 Combination between software and contribution

Potential Gaps There is a notable gap in research concerning the utilization of Prometheus in performance evaluation for cloud-native applications. Although many studies focus on methodologies or tools, more research is needed on leveraging Prometheus for performance assessment. Prometheus, with its monitoring capabilities, offers a promising approach for evaluating and optimizing the performance of cloud-native applications. By harnessing its metrics collection and querying functionalities, researchers could delve deeper into understanding the intricate interplay between application components and Kubernetes infrastructure, thus fostering more informed decision-making processes to enhance performance and scalability.

Prometheus offers numerous advantages in monitoring and managing metrics within various environments. It integrates with different metrics exporters, providing a view of application and system performance [PS29]. By connecting with Kubernetes' API and utilizing components such as node exporter and kube-state-metrics, Prometheus enables detailed monitoring of cluster nodes and Kubernetes

objects, enhancing system visibility and control [PS21]. Additionally, Prometheus can be extended to collect custom metrics from deployments, allowing fine-grained control over system scaling and performance optimization [PS25]. Furthermore, by supplementing real-time resource insights and facilitating the integration of tracing tools, Prometheus enhances data analysis capabilities and aids in identifying bottlenecks in large-scale applications. Within the domain of cloud-native applications, a prominent research gap lies in leveraging AWS tools to introduce novel methods. While existing studies have made strides in various aspects, such as tool development or empirical analyses, more methodological papers are still needed to explore the potential of AWS services in advancing cloud-native paradigms. AWS offers a diverse array of services, from container orchestration with Amazon ECS or EKS to serverless computing with AWS Lambda, providing a rich landscape for methodological frameworks. By elucidating new methodologies that harness the capabilities of AWS tools, researchers could unlock fresh avenues for optimizing performance, scalability, and resilience in cloud-native applications.

5.3 Objective x Main Research Contribution

Figure 9 presents the relationship between objective and main research contribution.

Prevalent Research Domains Papers that primarily contribute to performance evaluation often concentrate on orchestration in cloud-native applications. These studies assess the efficacy of various orchestration strategies within cloud-native environments, addressing their impact on performance metrics such as latency, throughput, and resource utilization. By meticulously analyzing orchestration techniques, such as those facilitated by Kubernetes, Docker Swarm, or Amazon ECS, these papers provide invaluable insights into the optimal deployment and management of containerized applications. As the research papers highlight, orchestration is familiar in cloud-native architectures to ensure high availability and low latency. The deployment of cloud-native edge slices emphasizes the need for efficient resource allocation and management to avoid over-provisioning and optimize performance [PS06]. Additionally, the performance evaluation of service recovery strategies in cloud-native radio access networks underscores the importance of orchestrating recovery mechanisms to maintain network reliability and minimize downtime [PS21]. Furthermore, the implementation of a high-performance software load balancer in cloud-native environments showcases the significance of orchestrating load distribution to enhance system availability and responsiveness [PS17].

Potential Gaps A notable research gap emerges in the evaluation of network performance within papers where the method serves as the primary contribution. While other types of contributions, such as tool development or empirical studies, have garnered attention, methodological papers specifically addressing network performance evaluation still need to be included. By focusing on this aspect, researchers can offer insights and practical solutions to address the unique challenges of distributed networking in the cloud, ultimately advancing the efficiency, reliability, and scalability of cloud-native applications.

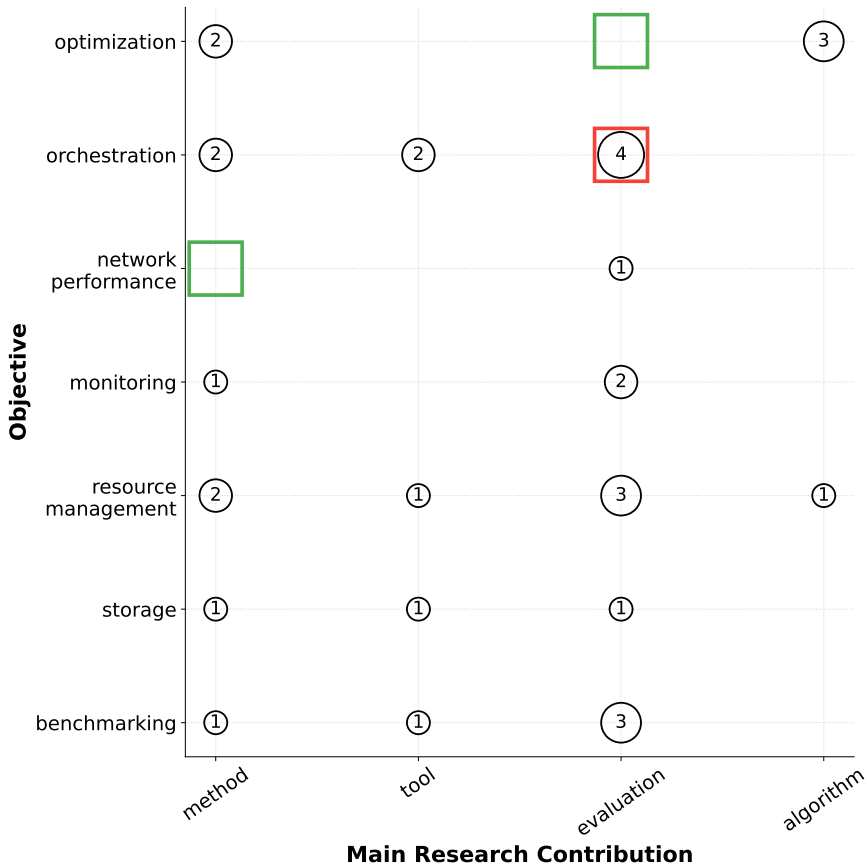


Fig. 9 Combination between objective and contribution

Another noticeable gap exists in research pertaining to optimization within papers where the evaluation itself serves as the primary contribution. While numerous studies focus on evaluating various aspects of cloud-native architectures, ranging from performance to scalability, more research still needs to be dedicated to optimization methodologies within this context. This gap presents an opportunity for exploration aimed at developing novel optimization techniques explicitly tailored to the intricacies of cloud-native environments. By addressing this gap, researchers can contribute insights and practical solutions to enhance the efficiency, resource utilization, and overall performance of cloud-native applications. Closing this research gap holds immense promise for driving advances in cloud-native computing.

5.4 Metrics x Objectives

Figure 10 presents the relationship between metrics and objectives.

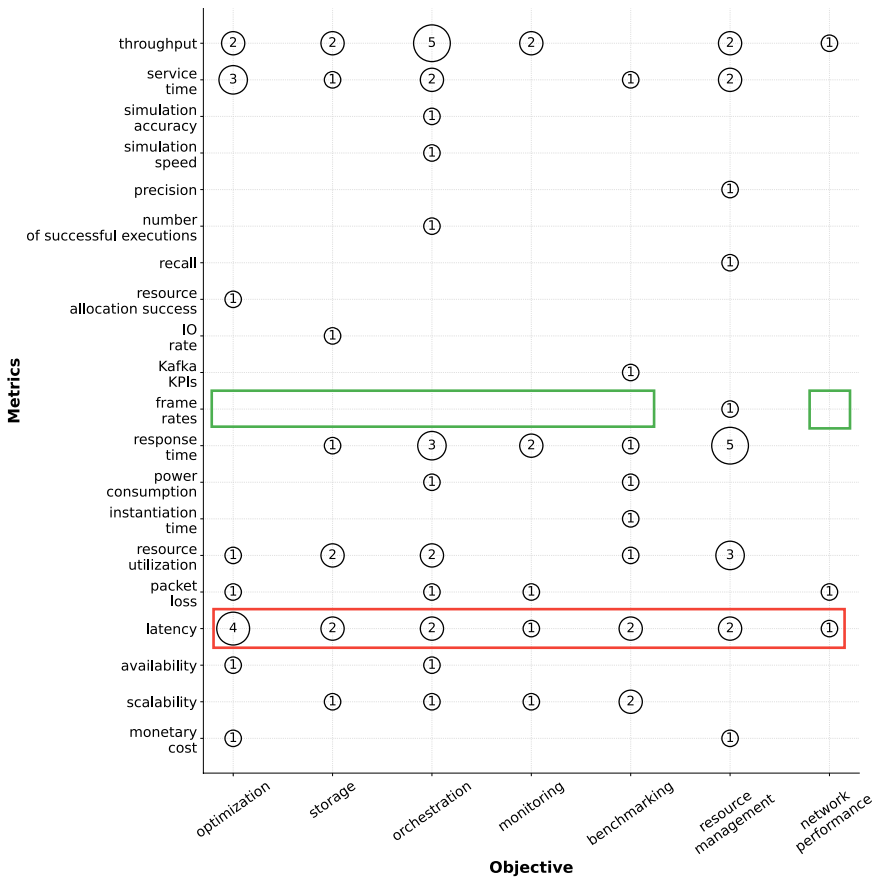


Fig. 10 Combination between metrics and objectives

Prevalent Research Domains Latency has been extensively explored across various objectives, with a predominant focus on optimization within the evaluation process. From enhancing performance to fine-tuning resource allocation, researchers have scrutinized latency metrics to glean insights into optimizing cloud-native applications for efficiency and responsiveness. By exploring latency, studies aim to streamline operations, minimize response times, and maximize user experience in distributed environments. This exploration underscores the critical role of latency optimization in the continual improvement of cloud-native ecosystems. Utilizing the latency metric is motivated by its critical role in driving the deployment of cloud-native microservices at the edge, as highlighted in [PS14]. Latency, a key performance indicator, encompasses communication latency among microservices and service latency, which includes inter-microservice latency and processing time, impacting overall functionality [PS11]. In the realm of telecommunication networks, latency, packet loss, and throughput are fundamental metrics for performance

evaluation, with delay representing the time taken for packet transmission across a network path [PS02]. Furthermore, in latency-sensitive applications, execution time and invocation delay influence the software, hardware capabilities, and network-specific parameters, highlighting the importance of measuring and optimizing latency for efficient operations [PS04].

Potential Gaps There exists a notable gap in research concerning the utilization of frame rate as a metric. While metrics like latency and throughput have garnered attention, frame rate, which is particularly relevant in applications involving multimedia content or real-time graphics, remains largely unexplored. This gap represents an opportunity for researchers to explore the impact of frame rate on user experience, system performance, and resource utilization within cloud-native environments. By incorporating frame rate analysis into evaluation methodologies, researchers can gain valuable insights into optimizing cloud-native applications for tasks such as video streaming, gaming, video calling, or virtual reality. Addressing this research gap holds promise for enhancing the responsiveness, visual quality, and overall effectiveness of cloud-native applications across a diverse range of use cases.

5.5 Software x Objective

Figure 11 presents the relationship between software and objectives.

Prevalent Research Domains Kubernetes remains a focal point, particularly in the domain of resource management. Researchers are exploring mechanisms of orchestrating and optimizing resource allocation within Kubernetes clusters. From fine-tuning scheduling algorithms to exploring ways of scaling resources dynamically, the pursuit is to enhance efficiency, resilience, and cost-effectiveness in containerized environments. This focus reflects Kubernetes' growing importance as the backbone of modern cloud-native architectures, driving continuous to meet the evolving demands of scalable and reliable applications. The platform orchestrates applications, enforces resource limits, and optimizes resource consumption through features like the Completely Fair Scheduler bandwidth control [PS10]. Research projects explore the performance of cloud-native frameworks, comparing platforms like Docker and Kubernetes in terms of completion time and resource utilization [PS20]. Additionally, Kubernetes offers various resources, such as Services, Ingresses, and ConfigMaps, for efficient deployment and management of components in different environments [PS09]. The use of Kubernetes as a container orchestrator simplifies the deployment and management of distributed systems like Elasticsearch, enhancing scalability and reliability in complex architectures [PS21]. Furthermore, Kubernetes is essential for managing containerized systems in cloud environments, ensuring efficient resource assignment and lifecycle management for application containers running in pods on cluster nodes [PS19].

Potential Gaps Despite the extensive utilization of TensorFlow, PyTorch, Spark, and Grafana in resource management objectives within cloud-native ecosystems, a conspicuous gap emerges in their application towards performance evaluation of cloud-native applications. While these tools excel in facilitating efficient resource allocation and monitoring within Kubernetes clusters, their potential for evaluating

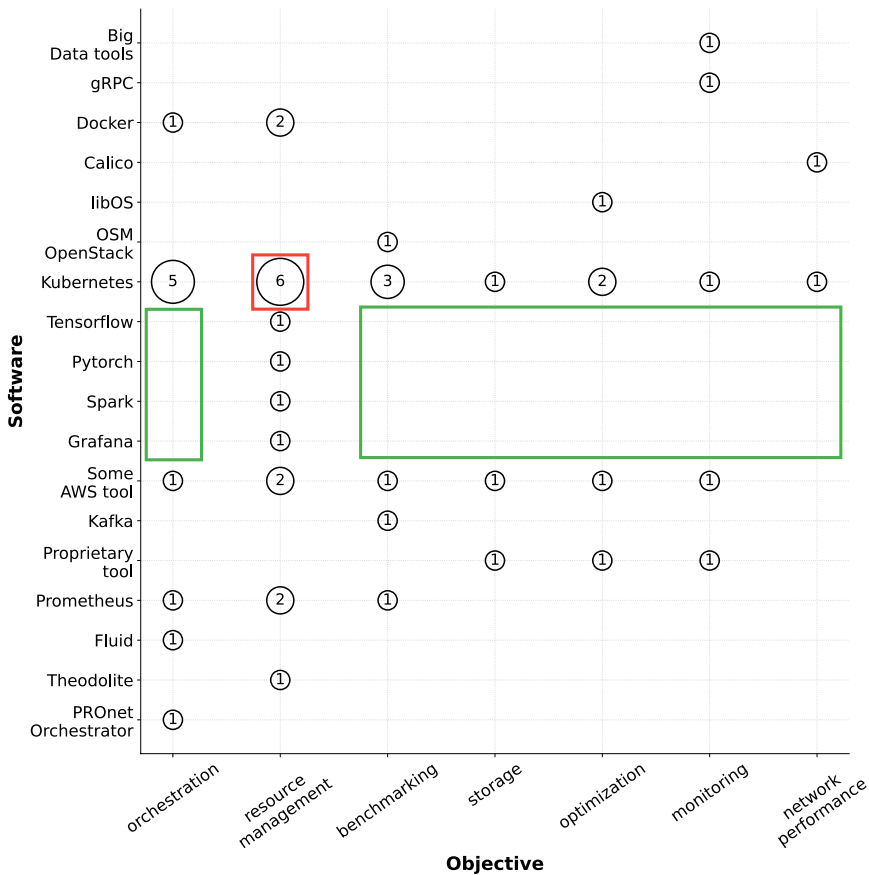


Fig. 11 Combination between software and objective

the performance of cloud-native applications still needs to be explored. Incorporating these frameworks into the evaluation process could offer invaluable insights into the runtime behavior, scalability, and optimization opportunities of cloud-native applications, enabling developers to make informed decisions to enhance their overall performance. Grafana, for example, is a powerful tool that finds motivation for its usage in various contexts. Grafana is utilized for monitoring and visualizing data in real time, making it valuable for analyzing complex systems and identifying trends efficiently.

5.6 Metric x Software

Figure 12 presents the relationship between metric and software.

Prevalent Research Domains The focus once again resides on AWS tools, with a particular emphasis on service time. Scholars and practitioners are deeply

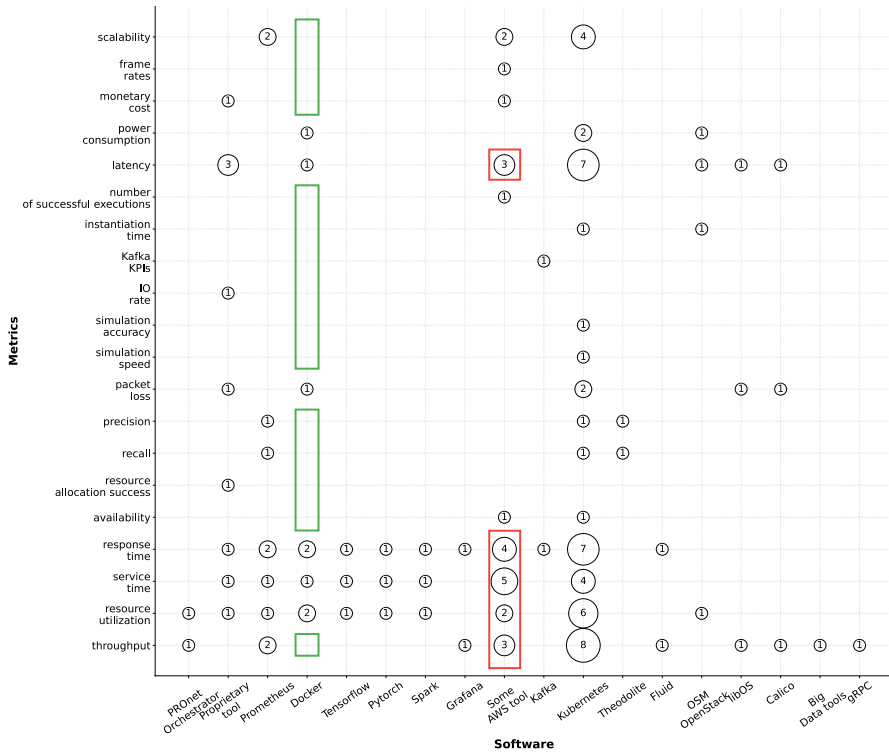


Fig. 12 Combination between metrics and software

engaged in exploring the nuances of service time metrics within the AWS ecosystem, aiming to refine and optimize the delivery of cloud services. From analyzing the latency of various AWS services to investigating strategies for minimizing service time fluctuations, the pursuit is directed toward enhancing the reliability and responsiveness of cloud-based applications. This renewed emphasis underscores the critical role that service time plays in shaping the user experience and operational efficiency within cloud environments, propelling ongoing efforts to innovate and refine AWS tools for optimal performance and service delivery.

Potential Gaps Despite the adoption of Docker in cloud-native environments, there exists a noticeable gap in its utilization for the performance evaluation of cloud-native applications. While Docker excels in containerization and deployment, its potential for performance assessment still needs to be explored. Current evaluation practices often focus on resource utilization metrics and container orchestration dynamics, overlooking deeper insights into application-level performance characteristics. Incorporating Docker into performance evaluation frameworks could offer valuable insights into runtime behavior, scalability, and optimization opportunities specific to containerized applications. Closing this gap presents an exciting opportunity for researchers and practitioners to enhance the effectiveness of performance evaluation methodologies in cloud-native

environments, ultimately leading to more efficient application deployments. For example, a potential metric that needs to be explored is throughput.

Docker is motivated by its high performance compared to competitors such as VMs and Unikernels, especially in reducing startup and serving time [PS07]. It is designed to deploy and run distributed applications efficiently without launching entire virtual machines. It is ideal for cloud-native applications and microservices managed on elastic infrastructure through agile processes [PS20]. Docker's containerization technology manages AI model training platforms with GPU hardware in cloud environments, optimizing resource utilization for data-hungry and computation-intensive workloads such as deep learning [PS06]. Furthermore, Docker's open-source containerization service, coupled with orchestration systems such as Kubernetes, allows efficient resource utilization in cloud environments, making it essential for implementing diverse services through containerized systems in the cloud [PS26].

6 Research Challenges and Directions

The performance evaluation of cloud-native applications presents several challenges that researchers and practitioners need to address. Some of the main challenges include:

Dynamic Nature Cloud-native applications, such as those leveraging technologies like Docker and Kubernetes, are dynamic. Components like containers and microservices can scale up or down based on demand, leading to a constantly changing application environment. This dynamic nature poses a challenge for performance evaluation, as traditional metrics may not accurately reflect real-time performance under varying loads. Cloud-native applications, leveraging technologies like Docker and Kubernetes, exhibit a dynamic nature, with components such as containers and microservices continuously scaling based on demand [PS20] [PS10]. This dynamism poses challenges for performance evaluation, as traditional metrics may need to capture real-time performance fluctuations under varying workloads effectively. Cloud-native platforms like Kubernetes allow for dynamic resource allocation, but issues like CPU throttling can impact overall system performance [PS05]. Additionally, in cloud-native network functions, the kernel space is identified as having more potential for network acceleration, especially in virtual machine deployments [PS23]. Understanding and adapting to this dynamic environment helps optimize the performance of cloud-native applications in response to changing demands and resource constraints.

Scalability Ensuring consistent performance as applications scale horizontally (adding more instances) or vertically (increasing resources per instance) is not just a nice to have but a necessity. Performance benchmarks need to be adaptable and account for these scaling scenarios to provide meaningful insights. Scalability is a fundamental requirement for cloud-native applications, which are specifically engineered to operate across dynamic and scalable cloud infrastructures [PS20]. Cloud-native tools such as Kubernetes play an essential role in enabling scalability by efficiently orchestrating application containers [PS18]. Leveraging

mechanisms like Objects Lambda and S3 Select can improve the performance of cloud-native distributed object storage. These mechanisms optimize data processing and movement closer to storage nodes, enhancing compute efficiency and performance gains [PS19]. Additionally, the scalability of cloud-native network functions faces challenges in network performance, particularly in packet processing delays. Solutions focus on accelerating processing efficiency in both the user and kernel spaces [PS12]. Overall, the scalability of cloud-native applications is a critical consideration in ensuring their ability to handle increasing workloads and demands in dynamic cloud environments.

Complexity of Ecosystem Cloud-native applications often leverage a variety of tools and technologies, such as container orchestration platforms (e.g., Kubernetes), service meshes, and serverless computing frameworks. Evaluating performance across this diverse ecosystem requires understanding and integrating different metrics and tools. The evaluation of performance in cloud-native applications is linked to the complexity of the ecosystem surrounding them. As highlighted in [PS30], navigating the intricate ecosystem surrounding cloud-native applications presents a challenging yet essential consideration. Factors such as the network infrastructure, deployment environments, integration with external services, and the overall architecture can impact the scalability, reliability, and resource utilization of cloud-native applications. Understanding and assessing this complexity is not only essential but also an exciting part of conducting accurate performance evaluations, identifying bottlenecks, optimizing resource allocation, and ensuring the efficient operation of cloud-native applications in diverse and dynamic environments.

Monitoring and Observability Because cloud-native applications are distributed, it becomes critical to monitor performance metrics across multiple components and services. Ensuring observability without impacting application performance itself is a challenge. The impact of monitoring and observability on the scalability of cloud-native applications is significant. In [PS19], it is highlighted that observability is vital for tracing application execution and detecting malfunctions and anomalies in cloud-based applications and IoT use cases. Observability refers to the ability to understand a system's internal state based on its external outputs. In the context of cloud-native applications, this means being able to trace the execution of various components and services, detect any malfunctions or anomalies, and identify the root causes of errors. Efficient monitoring and observability enable the identification of performance issues, bottlenecks, and resource demands, allowing for proactive scaling and optimization to meet the increasing workload demands. Monitoring and observability, by collecting and analyzing relevant data, contribute to enhancing the performance, resource utilization, and overall scalability of cloud-native applications.

Multi-Tenancy Issues In multi-tenant environments, where multiple users or applications share the same infrastructure resources, ensuring fair allocation of resources and evaluating performance in isolation from other tenants is critical. In the context of cloud-native applications, multi-tenancy refers to a cloud system's capability to support multiple users or tenants sharing the same resources and infrastructure. In a multi-tenant environment, ensuring equitable resource allocation, preventing performance bottlenecks, and maintaining data isolation are essential factors

that influence the overall performance of cloud-native applications. Issues such as noisy neighbors, where one tenant's high resource demands affect other tenants' performance, can arise and lead to performance fluctuations. Moreover, security concerns related to data privacy and integrity increase in multi-tenant environments, necessitating measures to ensure secure isolation and compliance with data regulations. Addressing multi-tenancy issues requires careful monitoring, workload management, and performance optimization strategies tailored to the unique requirements of cloud-native applications in shared environments [PS06].

Security and Compliance Performance evaluation must also consider security aspects such as data protection, access control, and compliance requirements (e.g., General Data Protection Regulation). Ensuring that performance benchmarks do not compromise security or regulatory compliance adds another layer of complexity. Security plays a critical role in the performance evaluation of cloud-native applications [PS03]. Security is not just a side note in the performance evaluation of cloud-native applications. As these applications are deployed in dynamic and shared environments, assessing their performance involves considerations beyond speed and efficiency. Security concerns such as data protection, access control, and threat detection directly impact the application's overall performance and reliability. Vulnerabilities in security measures can lead to breaches, data loss, or service interruptions, ultimately affecting the application's performance metrics and user experience. By integrating security protocols and conducting thorough security evaluations during performance assessments, organizations can ensure the safety and integrity of their cloud-native applications while optimizing their performance for enhanced operational efficiency and user trust.

Resource Utilization Optimization Optimizing resource utilization while maintaining performance levels is essential for cost-effectiveness and efficient operation of cloud-native applications. Performance evaluations need to balance resource efficiency with performance metrics. Efficient resource utilization is vital in evaluating the performance of cloud-native applications [PS29]. Efficient utilization of resources such as CPU, memory, and storage directly impacts the overall performance, scalability, and cost-effectiveness of applications running in cloud environments. Optimizing resource utilization ensures that the allocated resources are utilized effectively, minimizing wastage and enhancing the application's responsiveness and stability. By fine-tuning resource allocation based on workload demands, organizations can improve the application's efficiency, reduce latency, and enhance the user experience. Additionally, optimizing resource utilization contributes to better resource management, increased system reliability, and optimized cost efficiency in cloud-native environments. Overall, prioritizing resource utilization optimization is essential for maximizing the performance and reliability of cloud-native applications while effectively managing operational costs.

Heterogeneity of Workloads Cloud-native applications may handle a wide range of workloads, from batch processing to real-time data streaming. Evaluating performance across these heterogeneous workloads requires adaptable performance evaluation strategies. The diversity of workloads impacts the performance evaluation of cloud-native applications [PS25]. By encompassing a diverse range of workloads characterized by varying attributes, such as data intensity, metadata intensity,

read/write intensity, and specific application requirements. Organizations can gain understanding of their application's scalability, efficiency, and responsiveness under different operational scenarios. Evaluating cloud-native applications with heterogeneous workloads allows for a more holistic assessment of performance capabilities, enabling organizations to tailor their resource allocations, optimize system configurations, and enhance overall application robustness. This approach ensures that cloud-native applications are tested under realistic conditions, reflecting the dynamic nature of modern workloads in cloud environments. We did not explore this aspect in the mapping because the papers utilized a wide variety of workloads, which made categorization and analysis challenging. Therefore, this aspect could be explored further in future research endeavors.

7 Conclusion

This paper has provided a detailed exploration of key themes and insights derived from the individual analysis of research questions related to metrics, software programs, objectives, and primary research contributions in cloud-native performance evaluation. Through meticulous examination, we have highlighted trends and preferences within the literature. Metrics such as latency, response time, and throughput emerged as critical indicators of system performance, offering insights into network efficiency and user experience. Software programs like Kubernetes and AWS tools manages and optimizes cloud-native applications, underscoring their adoption and impact. Objectives such as resource management and orchestration were identified as primary focuses in enhancing scalability, reliability, and cost-effectiveness across cloud environments. Moreover, the emphasis on performance evaluation as a contribution underscores its importance in advancing methodologies and tools for assessing cloud-native application performance.

The paper also presented a multidimensional analysis across various criteria. By exploring intersections of metrics, objectives, and software tools, this study has identified both prevalent research domains and potential gaps for future exploration. The prevalence of throughput metrics alongside methodologies underscores their critical role in enhancing system performance and scalability. Conversely, gaps in research, such as the under-exploration of metrics like packet loss, availability, and monetary cost, highlight opportunities for developing novel evaluation methods. Gaps remain, particularly in leveraging tools like Prometheus for performance assessment and exploring Docker's potential in application-level performance evaluations.

This study identifies three central challenges to its validity: (i) the possibility that the research questions may not cover all relevant studies on cloud-native application performance, (ii) the risk of selection bias that could lead to the exclusion of pertinent research, and (iii) the subjective nature involved in extracting data. To mitigate these challenges, the study employed a research strategy and involved collaborative classification by multiple co-authors, and the paper project benefitted from external expert supervision. These efforts aimed to ensure a thorough exploration of the literature and enhance the reliability of the study's findings.

The exploration of cloud-native application performance evaluation presents several compelling avenues for future research. Future research may dive into the underexplored challenges related to cloud-native performance evaluation. The new research can focus on the research challenges presented in the last section. An essential focus is developing adaptive metrics and evaluation methodologies that accurately reflect the dynamic nature of cloud-native environments. Another promising direction is advancing monitoring and observability techniques tailored for cloud-native applications.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical Approval Not applicable

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Authors and Affiliations

Francisco Airton Silva¹ · Fernando A. M. Trinta² · Michel S. Bonfim² · José Antonio F. de Macedo² · Paulo A. L. Rego² · Vinícius Lagrota³

✉ Francisco Airton Silva
faps@ufpi.edu.br

Fernando A. M. Trinta
fernando.trinta@dc.ufc.br

Michel S. Bonfim
michelsb@ufc.br

José Antonio F. de Macedo
jose.macedo@dc.ufc.br

Paulo A. L. Rego
paulo@dc.ufc.br

Vinícius Lagrota
vinicius.1232@abin.gov.br

¹ Federal University of Piauí (UFPI), Picos, Piauí, Brazil

² Federal University of Ceará (UFC), Fortaleza, Ceará, Brazil

³ Research and Development Center for Communication Security (CEPESC), Brasília, Distrito Federal, Brazil